

Final Exam

Name: _____

Time: 2 hours

Instructions

Use a **pencil**. **Show all your work and explain your reasoning** — you are a mathematician, so a bare answer with no justification will not earn points. Partial credit will be given, so explain your thought process as much as possible, even if you can't answer it. When a question asks you to *explain in words*, write in full sentences. The test is worth 221 points. You have 2 hours; manage your time, and do not get stuck on any one problem. If something is unclear, ask Leo.

Part 1 — Speaking Math

1. [6 pts] Classify each of the following as an **expression**, an **equation**, or a **formula**.

(a) $4x$ _____

(b) $y = 5x$ (assume no context is given) _____

(c) $\text{Area} = \text{Width} \cdot \text{Height}$ _____

(d) $3x + 1 = 10$ _____

(e) 12 _____

In one sentence: what makes something a **formula** rather than an equation?

2. [10 pts] Consider the expression $x^2y + 8x - 4$.

(a) How many terms does it have?

Answer: _____

(b) Fill in the table.

Term	Coefficient	Variable(s)	Exponent(s)
x^2y			
$8x$			
-4			

(c) Mark each True or False. If false, **write the corrected version**.

T / F The coefficient of the x -term is 1.

T / F -4 is a term.

T / F x^2y and $8x$ are like terms.

3. [6 pts] Speaking math.

(a) Read this aloud and write down the complete English sentence you would say:

$$x(x - 3)^2 + 6$$

(b) Write the algebraic expression for: “the quantity four x minus two, over the quantity x plus three.”

Part 2 — Sets

Recall

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \quad (\text{the integers})$$

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\} \quad (\text{the natural numbers})$$

$$\mathbb{R} = \{\dots, -3.24, 0, 1.99, \sqrt{2}, \pi, 10, \dots\} \quad (\text{the real numbers})$$

4. [8 pts] Let $S = \{4, 8, 12, 16\}$.

(a) Fill in $\in$ or $\notin$: 8 ____ S 5 ____ S 16 ____ S 0 ____ S

(b) Write S in set-builder notation (you can use words for the condition).

(c) Write each in roster notation.

(i) $\{x \in \mathbb{Z} \mid -3 \leq x \leq 1\} =$ _____

(ii) $\{x \in \mathbb{N} \mid x^2 < 10\} =$ _____

5. [4 pts] Subsets.

(a) List **all** subsets of $\{p, q\}$ Which one is the **improper** subset?

6. [8 pts] Let $U = \{1, 2, 3, \dots, 10\}$, $A = \{1, 4, 6, 9\}$, $B = \{2, 4, 6, 8\}$. Fill out the following chart:

Expression	Result
$A \cup B$	
$A \cap B$	
$A \setminus B$	
A^c (in U)	
$ A $	
$ A \cup B $	

7. [4 pts] Real-world sets.

(a) At a café, 30 customers came in one morning: 16 ordered coffee, 13 ordered a pastry, and 6 ordered **both**. How many ordered **at least one** of the two? Show your work.

Part 3 — Functions

8. [6 pts] Let $f(x) = x^2 + 1$.

(a) Fill in the table.

x	0	2	-1
$f(x)$			

(b) Find x such that $f(x) = 7$. Answer: _____

(c) Write f as a set of **ordered pairs** for the inputs $\{0, 2, -1\}$.

9. [8 pts] For each rule, decide whether it is a valid function from A to B . That is, $f : A \rightarrow B$. If it is, state whether it is **injective** and whether it is **surjective**. If it is not a function, explain why not.

(a) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$: $1 \mapsto 1$, $2 \mapsto 3$, $3 \mapsto 5$.

(b) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$: $1 \mapsto 2$, $2 \mapsto 4$, $1 \mapsto 3$, $3 \mapsto 1$.

(c) $A = \{1, 2, 3\}$, $B = \{x, y\}$: $1 \mapsto x$, $2 \mapsto y$, $3 \mapsto y$.

(d) $A = \{1, 2, 3\}$, $B = \{a, b, c\}$: $1 \mapsto b$, $2 \mapsto a$.

10. [4 pts] Domain, codomain, range.

For $g: \mathbb{N} \rightarrow \mathbb{N}$ defined by $g(x) = x^2 + 1$, state the domain, the codomain, and the range (no need to show rigorously, just give the answer).

11. [8 pts] Injective, surjective, bijective.

(a) Write a function $\{1, 2, 3\} \rightarrow \{a, b, c\}$ that is **bijective**.

(b) Write a function $\mathbb{Z} \rightarrow \mathbb{Z}$ that is **injective but not surjective**.

12. [8 pts] Formally prove that $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 2$ is bijective.

13. [6 pts] Formally prove that $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^2$ is **not** bijective.

14. [6 pts] Real-world application. An arcade charges \$7 to get in, plus \$3 for each game token you buy. You buy t game tokens.

(a) Write a function $C(t)$ for the total cost (in dollars) of your visit. Give its signature ($C: A \rightarrow B$) and its rule.

(b) In one sentence, what would it mean **in this context** for C to be injective?

(c) Is C surjective onto all positive integers?

15. [8 pts] Big ideas about infinity. Answer each in words.

(a) We cannot finish counting an infinite set. So how do we decide whether two infinite sets have the **same size**?

(b) What makes the **Collatz Conjecture** a conjecture, but not a theorem?

16. [10 pts] Determine if $\mathbb{N}$ and $\mathbb{Z}$ are the same size. Explain your answer.

Domain & Range

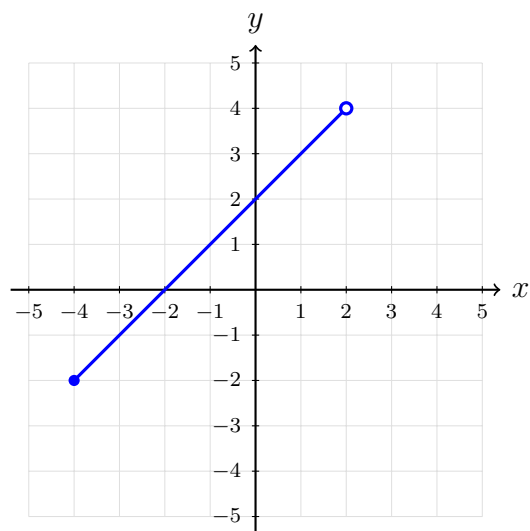
17. [14 pts] Find the **domain** and **range** of each function. Rigorously show your reasoning.

(a) $f(x) = \sqrt{x - 3}$

(b) $g(x) = \frac{4x}{x-1}$

18. [14 pts] Find the **domain** of the function $f(x) = \frac{\sqrt{x-3}}{x-1}$.

19. [8 pts] The graph of a function p is shown below (the filled dot is included, the open dot is not). State its **domain** and **range** using interval notation.

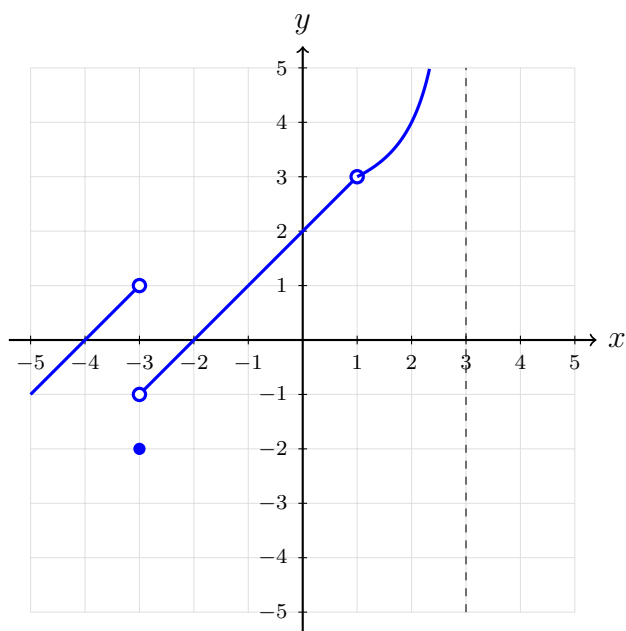


Domain: _____

Range: _____

Limits — Reading, Tables & Concepts

20. [16 pts] Use the graph of H below (the dashed line is a vertical asymptote). Open circles are missing points; filled circles are actual values.



(a) $\lim_{x \rightarrow -3^-} H(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow -3^+} H(x) = \underline{\hspace{2cm}}$ (c) $\lim_{x \rightarrow -3} H(x) = \underline{\hspace{2cm}}$

(d) $H(-3) = \underline{\hspace{2cm}}$ (e) $\lim_{x \rightarrow 1} H(x) = \underline{\hspace{2cm}}$ (f) Is $H(1)$ defined? $\underline{\hspace{2cm}}$

(g) $\lim_{x \rightarrow 3^-} H(x) = \underline{\hspace{2cm}}$

21. [6 pts] The table shows values of $f(x) = \frac{x^2 - 4}{x - 2}$ near $x = 2$.

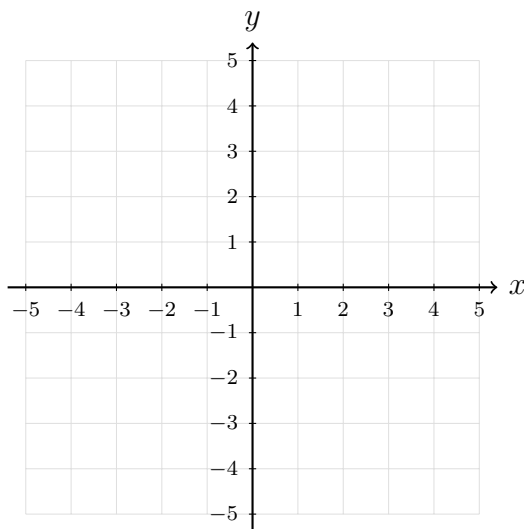
x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	3.9	3.99	3.999	4.001	4.01	4.1

- (a) $\lim_{x \rightarrow 2} f(x) =$ _____
- (b) Is $f(2)$ defined? Why or why not?

22. [8 pts] In your own words:

- (a) Explain the difference between saying “ $\lim_{x \rightarrow a} f(x)$ does not exist” and saying “ $f(a)$ is undefined.”

- (b) Describe or sketch a function for which the limit at a point *exists* even though the function is *undefined* there.



Finding Limits with Algebra

23. [22 pts] Evaluate each limit. Show your steps.

(a) [3] $\lim_{x \rightarrow 2} (x^2 - 3x + 1)$

(b) [4] $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

(c) [4] $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2}$

(d) [4] For $f(x) = \begin{cases} x + 4 & x < 2 \\ x^2 - 1 & x \geq 2 \end{cases}$, find $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$, and $\lim_{x \rightarrow 2} f(x)$.

(e) [4] $\lim_{x \rightarrow 2^+} \frac{1}{x - 2}$ and $\lim_{x \rightarrow 2^-} \frac{1}{x - 2}$. Does $\lim_{x \rightarrow 2} \frac{1}{x - 2}$ exist?

Limits at Infinity (End Behavior)

24. [12 pts] Find each limit using the dominant-term shortcut. State whether the answer is a number, 0, $+\infty$, or $-\infty$.

(a) $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x}{2x^3 + 7}$

(b) $\lim_{x \rightarrow \infty} \frac{3x + 1}{x^2 - 4}$

(c) $\lim_{x \rightarrow \infty} \frac{x^2 + 5}{4x + 1}$

(d) $\lim_{x \rightarrow \infty} \frac{6x^2 - x}{2x^2 + 3x - 5}$

25. [14 pts] Now do it rigorously: divide the top and bottom by the highest power of x , and use the fact that $\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$. For each step state which limit law you are using.

$$\lim_{x \rightarrow \infty} \frac{2x^2 - x}{5x^2 + 4}$$

26. What was the hardest part of our math classes? Do you believe one day in the future you'll be able to conquer it?